1. Principles of Remaining Useful Life (RUL) Prediction in Turbomachinery

Abstract: Accurate Remaining Useful Life (RUL) estimation is essential for condition-based maintenance and zero-unplanned-downtime manufacturing.

1.1 Prognostic Modeling Framework:
Prognostics models the transformation of healthy equipment into failed equipment over time.
Let X_t = [s_1(t), s_2(t), ..., s_p(t)] represent the multi-channel sensor observation vector at cycle t.
The health index H(t) in [0, 1] tracks continuous degradation:
H(t) = 1.0 - integral(dD/dt * dt, from 0 to t)

1.2 Attention-CNN-BiLSTM Architecture for Multi-Regime RUL Estimation:
Recent breakthroughs in deep learning combine 1D-CNNs for local feature extraction with Bidirectional LSTMs for long-range temporal degradation context.
Self-attention pooling dynamically weights critical degradation inflection cycles, resulting in superior performance on NASA C-MAPSS FD002 and FD004 subsets.
Validation benchmarks demonstrate an RMSE drop to 27.08 cycles on FD002 compared to 42.1 cycles for standard Multi-Layer Perceptrons.

2. Degradation Inflection and Paris' Crack Propagation Mechanics

2.1 Paris-Erdogan Fatigue Crack Growth Law:
Sub-surface rolling contact fatigue (RCF) in bearings and compressor blade roots follows:
da/dN = C * (Delta K)^m
where 'a' is crack depth, 'N' is cycles, 'Delta K' is stress intensity factor range, and C, m are material constants.

2.2 Practical Maintenance Implications:
- Phase I (Micro-crack initiation): Sensor telemetry remains indistinguishable from healthy baseline (cycles 0 to ~100).
- Phase II (Stable crack growth): High-frequency vibration and slight temperature drift emerge (cycles 100 to ~160).
- Phase III (Accelerated propagation): Severe vibration, rapid EGT escalation, and acute failure risk (cycles 160 to failure).
Maintenance action MUST be scheduled before Phase III begins.